Simplex Tableau Method

Tuesday, August 12, 2025 10:05 PM

Objective function < cost func. (min) / profit func. (max)

$$\text{Max } 7x_1 + 6x_2 = z$$

$$\text{s.t. } 2x_1 + 4x_2 \le 16$$

$$3x_1 + 2x_2 \le 12$$

$$x_1, x_2 \ge 0$$ ← non-zero constraints

slack variables

$$2x_1 + 4x_2 + s_1 = 16$$

$$3x_1 + 2x_2 + s_2 = 12$$

$$x_1, x_2, s_1, s_2 \ge 0$$

Initial Tableau

Basis	C_B	x_1 7	x_2 6	s_1 0	s_2 0	b	min ratio
s_1	0	2	4	1	0	16	$16/2=8$
s_2	0	3	2	0	1	12	$12/3=4$ ← minimum
z_j		0	0	0	0	0	
$c_j - z_j$		7	6	0	0		

min row

Intersection

$0\times2 + 0\times3$ $\quad 0\times4 + 0\times2$ $\quad 0\times1 + 0\times0$ $\quad 0\times0 + 0\times1$ $\quad 0\times16 + 0\times12$

Basis	C_B	x_1 7	x_2 6	s_1 0	s_2 0	b	min ratio
s_1	0	0	8/3	1	-2/3	8	$8\times\frac{3}{8}=3$
x_1	7	1	2/3	0	1/3	4	$4\times\frac{3}{2}=6$
z_j		7	14/3	0	7/3	28	
$c_j - z_j$		0	4/3	0	-7/3		

R_1 old $- 2R_2$ new $\Rightarrow R_1$ new

x_1: $0\times0 + 7\times1 = 7$

x_2: $0\times8/3 + 7\times2/3 = 14/3$

s_1: $0\times1 + 7\times0 = 0$

s_2: $0\times-2/3 + 7\times1/3 = 7/3$

b: $0\times8 + 7\times4 = 28$

R_2 old $- \frac{2}{3} R_1$ new $\Rightarrow R_2$ new

$0 - \frac{2}{3}(1) = 1$ $\quad 4 - \frac{2}{3}(3) = 2$

$\frac{2}{3} - \frac{2}{3}(1) = 0$

$0 - \frac{2}{3}\left(\frac{3}{8}\right) = \frac{-1}{4}$

$1/3 - \frac{2}{3}\left(\frac{-1}{4}\right) = \frac{1}{2}$

Basis	C_B	x_1 7	x_2 6	s_1 0	s_2 0	b
x_2	6	0	1	3/8	-1/4	3
x_1	7	1	0	-1/4	1/2	2
z_j		7	6	1/2	2	32
$c_j - z_j$		0	0	-1/2	-2	

all is negative

$$\boxed{\begin{aligned} x_1 &= 2 \\ x_2 &= 3 \\ z &= 32 \end{aligned}}$$